Chapter 2 — Concept System and Information-Model Foundations

Purpose and scope

Defines semantic types, entity and record distinctions, relationships, states, events, claims, constraints, and machine-readable representations.

Source authority

Book I: Book I Articles II, IV, IX, and XI
Book II: Book II Chapters 01, 02, 22, 23, 24, 25, and 30
Book III: Book III Chapters 03, 04, 06, and 09

Normative semantic rules

• Every modeled concept MUST be classified as an entity, value, record, role, relationship, event, state, process, constraint, assessment, or governed decision when applicable.

• An Entity MUST remain distinguishable from its Identifier, records, attributes, roles, and representations.

• Every governed relationship MUST declare direction, source type, target type, cardinality, constraints, and lifecycle when material.

• Machine-readable Book X artifacts MUST preserve the same IDs and meanings as the canonical human-readable edition.

Canonical terms

HAL-TERM-0011 — Entity
A distinguishable thing with identity and continuity relevant to the HAL domain.
Distinction: An Entity is not the same as its record, identifier, state, role, or representation.
Type/category/status: Semantic type / Information model / Approved
Aliases: None
Source basis: Derived semantic synthesis
Book I: Book I Decisions 26, 29, 40, 47, and 51
Book II: Book II Chapters 04, 22, 23, 24, 25, and 30
Book III: Book III Chapters 3 and 4
Example: A dependent artifact cites `HAL-TERM-0011` when it uses **Entity** with this exact governed meaning: A distinguishable thing with identity and continuity relevant to the HAL domain.
Counterexample: A dependent artifact uses **Entity** in a way that violates its required distinction: An Entity is not the same as its record, identifier, state, role, or representation.
Relationships: None registered
Lifecycle: None registered
Constraint: An Entity is not the same as its record, identifier, state, role, or representation.

HAL-TERM-0012 — Value Object
An immutable value defined by its attributes rather than by independent identity.
Distinction: Changing a Value Object produces another value; it does not mutate an enduring identity.
Type/category/status: Semantic type / Information model / Approved
Aliases: None
Source basis: Derived semantic synthesis
Book I: Book I Decisions 26, 29, 40, 47, and 51
Book II: Book II Chapters 04, 22, 23, 24, 25, and 30
Book III: Book III Chapters 3 and 4
Example: A dependent artifact cites `HAL-TERM-0012` when it uses **Value Object** with this exact governed meaning: An immutable value defined by its attributes rather than by independent identity.
Counterexample: A dependent artifact uses **Value Object** in a way that violates its required distinction: Changing a Value Object produces another value; it does not mutate an enduring identity.
Relationships: None registered

Lifecycle: None registered
Constraint: Changing a Value Object produces another value; it does not mutate an enduring identity.

HAL-TERM-0013 — Record
A governed representation of facts, state, decisions, or observations retained by an owning domain.
Distinction: A Record is not necessarily authoritative evidence and is not interchangeable with the entity represented.
Type/category/status: Semantic type / Information model / Approved
Aliases: None
Source basis: Derived semantic synthesis
Book I: Book I Decisions 26, 29, 40, 47, and 51
Book II: Book II Chapters 04, 22, 23, 24, 25, and 30
Book III: Book III Chapters 3 and 4
Example: A dependent artifact cites `HAL-TERM-0013` when it uses **Record** with this exact governed meaning: A governed representation of facts, state, decisions, or observations retained by an owning domain.
Counterexample: A dependent artifact uses **Record** in a way that violates its required distinction: A Record is not necessarily authoritative evidence and is not interchangeable with the entity represented.
Relationships: None registered
Lifecycle: None registered
Constraint: A Record is not necessarily authoritative evidence and is not interchangeable with the entity represented.

HAL-TERM-0014 — Authoritative Record
The record owned by the designated source of truth for a governed state domain.
Distinction: A replica, cache, index, projection, or local copy is not authoritative merely because it is current.
Type/category/status: State role / Information model / Approved
Aliases: source of truth
Source basis: Derived semantic synthesis
Book I: Book I Decisions 26, 29, 40, 47, and 51
Book II: Book II Chapters 04, 22, 23, 24, 25, and 30
Book III: Book III Chapters 3 and 4
Example: A dependent artifact cites `HAL-TERM-0014` when it uses **Authoritative Record** with this exact governed meaning: The record owned by the designated source of truth for a governed state domain.
Counterexample: A dependent artifact uses **Authoritative Record** in a way that violates its required distinction: A replica, cache, index, projection, or local copy is not authoritative merely because it is current.
Relationships: None registered
Lifecycle: None registered
Constraint: A replica, cache, index, projection, or local copy is not authoritative merely because it is current.

HAL-TERM-0015 — Relationship
A typed association between concepts or entity instances with declared direction, cardinality, constraints, and lifecycle.
Distinction: Proximity or co-occurrence does not imply a governed Relationship.
Type/category/status: Semantic type / Information model / Approved
Aliases: None
Source basis: Derived semantic synthesis
Book I: Book I Decisions 26, 29, 40, 47, and 51
Book II: Book II Chapters 04, 22, 23, 24, 25, and 30
Book III: Book III Chapters 3 and 4
Example: A dependent artifact cites `HAL-TERM-0015` when it uses **Relationship** with this exact governed meaning: A typed association between concepts or entity instances with declared direction, cardinality, constraints, and lifecycle.
Counterexample: A dependent artifact uses **Relationship** in a way that violates its required distinction: Proximity or co-occurrence does not imply a governed Relationship.
Relationships: None registered
Lifecycle: None registered
Constraint: Proximity or co-occurrence does not imply a governed Relationship.

HAL-TERM-0016 — State
The values and lifecycle condition of an entity or process at a defined observation point.
Distinction: State is not an Event; an Event records a completed fact about change.
Type/category/status: Semantic type / Information model / Approved
Aliases: None
Source basis: Derived semantic synthesis
Book I: Book I Decisions 26, 29, 40, 47, and 51

Book II: Book II Chapters 04, 22, 23, 24, 25, and 30
Book III: Book III Chapters 3 and 4
Example: A dependent artifact cites `HAL-TERM-0016` when it uses **State** with this exact governed meaning: The values and lifecycle condition of an entity or process at a defined observation point.
Counterexample: A dependent artifact uses **State** in a way that violates its required distinction: State is not an Event; an Event records a completed fact about change.
Relationships: None registered
Lifecycle: None registered
Constraint: State is not an Event; an Event records a completed fact about change.

HAL-TERM-0017 — Lifecycle
The allowed states, transitions, entry conditions, exit conditions, terminal conditions, and evidence for a governed concept.
Distinction: A list of statuses without transition rules is not a complete Lifecycle.
Type/category/status: Semantic model / Information model / Approved
Aliases: None
Source basis: Derived semantic synthesis
Book I: Book I Decisions 26, 29, 40, 47, and 51
Book II: Book II Chapters 04, 22, 23, 24, 25, and 30
Book III: Book III Chapters 3 and 4
Example: A dependent artifact cites `HAL-TERM-0017` when it uses **Lifecycle** with this exact governed meaning: The allowed states, transitions, entry conditions, exit conditions, terminal conditions, and evidence for a governed concept.
Counterexample: A dependent artifact uses **Lifecycle** in a way that violates its required distinction: A list of statuses without transition rules is not a complete Lifecycle.
Relationships: None registered
Lifecycle: None registered
Constraint: A list of statuses without transition rules is not a complete Lifecycle.

HAL-TERM-0018 — Invariant
A condition required to remain true throughout a defined scope or transition set.
Distinction: An engineering invariant is not automatically a Constitutional invariant.
Type/category/status: Constraint / Information model / Approved
Aliases: None
Source basis: Derived semantic synthesis
Book I: Book I Decisions 26, 29, 40, 47, and 51
Book II: Book II Chapters 04, 22, 23, 24, 25, and 30
Book III: Book III Chapters 3 and 4
Example: A dependent artifact cites `HAL-TERM-0018` when it uses **Invariant** with this exact governed meaning: A condition required to remain true throughout a defined scope or transition set.
Counterexample: A dependent artifact uses **Invariant** in a way that violates its required distinction: An engineering invariant is not automatically a Constitutional invariant.
Relationships: None registered
Lifecycle: None registered
Constraint: An engineering invariant is not automatically a Constitutional invariant.

HAL-TERM-0019 — Claim
A proposition stated for evaluation and linked to its subject, issuer, scope, time, and supporting or opposing evidence.
Distinction: A Claim is not true merely because it is recorded or signed.
Type/category/status: Evidence-bearing assertion / Information model / Approved
Aliases: None
Source basis: Direct source normalization
Book I: Book I Decision 26
Book II: Book II Chapters 18 and 25
Book III: Book III Chapters 4, 5, 6, and 8
Example: A dependent artifact cites `HAL-TERM-0019` when it uses **Claim** with this exact governed meaning: A proposition stated for evaluation and linked to its subject, issuer, scope, time, and supporting or opposing evidence.
Counterexample: A dependent artifact uses **Claim** in a way that violates its required distinction: A Claim is not true merely because it is recorded or signed.
Relationships: HAL-REL-0027, HAL-REL-0029, HAL-REL-0030
Lifecycle: None registered
Constraint: A Claim is not true merely because it is recorded or signed.

HAL-TERM-0020 — Constraint
A condition that limits valid state, relationships, transitions, or behavior within a defined scope.
Distinction: A preference or target is not a Constraint unless its governing source makes it binding.
Type/category/status: Rule element / Information model / Approved
Aliases: None
Source basis: Derived semantic synthesis
Book I: Book I Decisions 26, 29, 40, 47, and 51
Book II: Book II Chapters 04, 22, 23, 24, 25, and 30
Book III: Book III Chapters 3 and 4
Example: A dependent artifact cites `HAL-TERM-0020` when it uses **Constraint** with this exact governed meaning: A condition that limits valid state, relationships, transitions, or behavior within a defined scope.
Counterexample: A dependent artifact uses **Constraint** in a way that violates its required distinction: A preference or target is not a Constraint unless its governing source makes it binding.
Relationships: None registered
Lifecycle: None registered
Constraint: A preference or target is not a Constraint unless its governing source makes it binding.

Relationship and lifecycle rules

Relationships involving these terms MUST use the typed relationship register. Lifecycle-bearing concepts MUST use the governed transition register; a status label alone never authorizes a prohibited transition.

Term-specific semantic evidence

Every Term Record above contains a governed example, counterexample, constraint, relationship reference set, and lifecycle reference set. “None registered” is explicit.

Anti-patterns

Semantic drift, authority laundering, and collapsing an entity with its identifier, record, attributes, role, or state are prohibited.

Verification

Verify ID and label uniqueness, precise source traceability, source-basis classification, relationship and lifecycle consistency, ambiguity controls, machine-readable parity, and cross-format content parity.

Change and deprecation

Every semantic change requires authority, compatibility, dependency impact, migration, reviewer, effective version, and any replacement and sunset conditions.

Review findings

Constitutional, architecture, engineering, semantic, usability, parity, and Owner-threshold reviews passed. No Owner Review item.